

A Study on the Rain Attenuation Characteristics Using Multiple Ku-Band Satellite Signals in Relation to Rain Area Motion

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SUMMARY Characteristics of rain attenuation statistics for the Ku-band satellite signals are investigated among the three earth stations at Osaka Electro-Communication University (OECU, Neyagawa, Osaka), NTT Yokosuka R&D Center (Yokosuka, Kanagawa), and satellite base station (Matsuyama, Ehime), respectively, from April 2022 to March 2023. The time difference of the attenuation occurrence among these stations is well explained by the motion of rain fronts and extratropical cyclones obtained from the weather charts. Rain attenuation characteristics such as duration time are shown to be largely affected by the speed of the rain area motion around each station. The scale of rain cells inferred from duration time and rain area speed is found to be increased up to around 10 km for the 5 dB attenuation according to the rainfall rate at each rainfall event. The time delayed diversity effects are also examined using the attenuation data observed at 1 min interval. The results are converted to the site diversity effects with the distance up to about 10 km by the rain area motion around each station. A novel method is thus proposed to estimate the site diversity effects from the 1 min attenuation data observed at only one station during at least one year. The joint time percentages agree fairly well with the ITU-R recommendations up to about 10 km distance at the original time percentages of more than 0.05%.

key words: satellite communications, rain attenuation, Ku-band, ITU-R recommendations, time delayed diversity, site diversity

1. Introduction

In future Beyond 5G/6G mobile communication systems, the development of network systems on a global and space scale is expected. In these systems, the non-terrestrial networks (NTN) composed of geostationary satellites (GEO), low earth orbit satellites (LEO), and high altitude platform station (HAPS) are considered to play an important role [1]. The non-terrestrial networks, however, need to comply with the terrestrial networks of Gbps or higher. Hence, the radio wave frequencies must range from the Ku band (14/12 GHz) to the Ka band (30/20 GHz) or higher. In this context, the rain attenuation that occurs in the frequency of above 10 GHz [2]–[4] will become a serious problem. Therefore, in the non-terrestrial network (NTN) using GEO/LEO/HAPS, the countermeasures against the rain attenuation will be very important in the future.

Recently, the rain attenuation of Ku-band JCSAT-5A beacon signal (12.75 GHz) is measured at NTT Yokosuka

R&D Center (Yokosuka City) and satellite base station (Matsuyama City) from April 2022 to March 2023 [5]. Also, the rain attenuation of Ku-band BS radio signal (11.84 GHz) have been measured at Osaka Electro-Communication University (OECU, Neyagawa City) for the long time from before [3]. In this study, the time difference of the attenuation occurrence is first investigated in light of rain area motion among these stations. In addition, rain attenuation characteristics such as duration time [2] are examined in more detail, compared to the motion of rain fronts and extratropical cyclones obtained from the weather charts. Then, the horizontal scale of rain cells accompanied by the attenuation exceeding 5 dB is discussed. Finally, based on the rain attenuation obtained at 1 min interval during one year, time delayed diversity effects are examined. Using rain area motion obtained from weather charts, we propose a novel method to convert the time delayed diversity effects to the site diversity effects around the station with the distance up to about 10 km.

Note that the site diversity effects were observed by six universities all over Japan from Hokkaido to Kyusyu, using JCSAT-1 in relatively long distances from 10 to 1400 km [6]. Also, the relationship between the alignment of the two station conducting site diversity and the direction of radio wave coming from the satellite was investigated, using the radar data of rainfall intensity [7]. On the other hand, as regards the time delayed diversity, the effects of time difference in 1 hour or more were studied for the Ka-band (21 GHz) broadcasting satellite [8]. In this study, the time delayed diversity effects are examined in a shorter time range of 1 to 60 min, and the relation to the site diversity effects is discussed in the distance up to 10 km.

2. Observation Systems

At Osaka Electro-Communication University (OECU), the Ku-band BS broadcast signal (11.84 GHz) is received by an offset parabolic antenna with a diameter of 1.2 m. The output signal of the IF level meter is recorded by a personal computer (PC) equipped with 16 bit AD converter at 1 sec interval. The rain attenuation is evaluated using the 1-min average of these received signal levels, together with a 1-min rain gauge with the resolution of 0.1 mm.

At NTT Yokosuka R&D Center (Yokosuka City) and satellite base station (Matsuyama City), on the other hand, the Ku-band JCSAT-5A beacon signal (12.75 GHz) is received by a backfire parabolic antenna with a diameter of 0.6 m (Yokosuka) and a Cassegrain antenna with a diame-

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ter of 4.5 m (Matsuyama), respectively. The beacon signal is sampled and recorded using spectrum analyzers at 20-second intervals [5]. In addition, rainfall rate is measured simultaneously by 1-min rain gauges with the resolution of 0.5 mm at both stations.

In this experiment, the signal levels measured by the two different satellites should be compared and discussed on the rain attenuation characteristics. The amount of rain attenuation depends on frequency and polarization of the radio wave, as well as elevation angle of the satellite propagation path. Consequently, the BS signal rain attenuation measured at Neyagawa station is converted to the JCSAT-5A rain attenuation, in accordance with that measured at Yokosuka and Matsuyama stations [9].

3. Lag Time of Rain Attenuation Occurrence between the Stations

Figure 1(a) shows the rain attenuation of Ku-band JCSAT-5A beacon signal measured at Matsuyama (blue) and Neyagawa (green) stations, respectively, on April 24, 2022. From Fig. 1(a), the rain attenuation is found to occur about 3 hours earlier in Matsuyama than Neyagawa. Also, Fig. 1(b) shows the cross-correlation function calculated between Matsuyama and Neyagawa stations. It is found that the lag time of attenuation between Matsuyama and Neyagawa is 3:24 with the cross-correlation coefficient of about 0.56.

Similarly, Fig. 2(a) shows the rain attenuation of Ku-band JCSAT-5A measured at Neyagawa (green) and Yokosuka (red) stations, respectively, on April 24, 2022. From Fig. 2(a), the rain attenuation is found to occur about 5 hours earlier in Neyagawa than Yokosuka. In addition, Fig. 2(b) shows the cross-correlation function calculated between Neyagawa and Yokosuka stations. It is found that the lag time of attenuation between Neyagawa and Yokosuka is 5:19 with the cross-correlation coefficients of about 0.56.

On this day, an extratropical cyclone (a low pressure, hereinafter simply referred to as a cyclone) proceeded from west to east off the coast of Pacific Ocean along the stationary front. The passage speed of the cyclone is, on an average, 18.3 m/s from the weather chart, with passage direction of about 80 deg clockwise from north over the three stations.

Figure 3 illustrates the passage length of the rain area projected to the two stations ($\overline{AB} \cos \theta$), where θ is the angle between the alignment of the stations A and B (blue line) and the passage direction (red arrow). Figure 3(a) is the case of cyclones along stationary fronts and without fronts, while Fig. 3(b) is the case of warm and cold fronts. Note that the speed and direction of (a) cyclones are estimated from the motion of their center. On the other hand, those of (b) warm and cold fronts are calculated from the motion perpendicular to the front lines [10].

The passage time projected to the direction between Matsuyama and Neyagawa (279 km, 68.6 deg from north) is estimated to be 4:11, and the passage time between Neyagawa and Yokosuka (372 km, 82.0 deg from north) is estimated to be 5:41, respectively, using the passage speed and

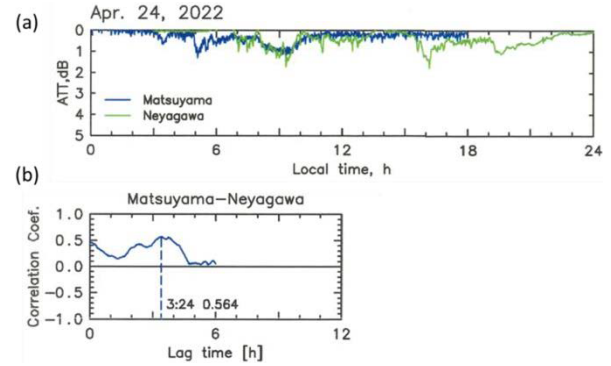


Fig. 1 (a) Rain attenuation of Ku-band JCSAT-5A beacon signal measured at Matsuyama (blue) and Neyagawa (green) stations, and (b) the cross-correlation function between the two stations.

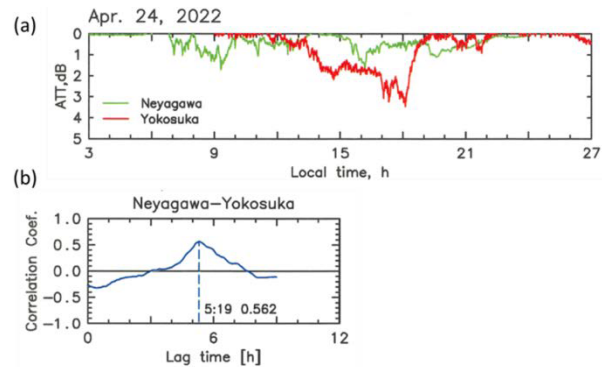


Fig. 2 (a) Rain attenuation of Ku-band JCSAT-5A beacon signal measured at Neyagawa (green) and Yokosuka (red) stations and (b) the cross-correlation function between the two stations.

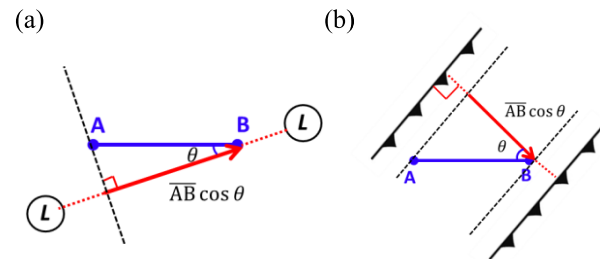


Fig. 3 Passage length of (a) cyclone and (b) rain front projected to the two stations ($\overline{AB} \cos \theta$), where θ is the angle between the alignment of the stations A and B and the passage direction.

Table 1 Lag time of attenuation and passage time of cyclone (hh:mm).

	Matsuyama-Neyagawa	Neyagawa-Yokosuka
Cross-correlation	3:24	5:19
Cyclone passage	4:11	5:41

direction of cyclone (18.3 m/s, 80 deg from north). These passage times agree well with the lag times obtained from the cross-correlation function in Figs. 1 and 2. The results are summarized in Table 1.

Similar results are obtained for 21 rainfall events from April 2022 to March 2023, in which the rainfall rate of more than 10 mm/h is measured for longer than 30 min at all of

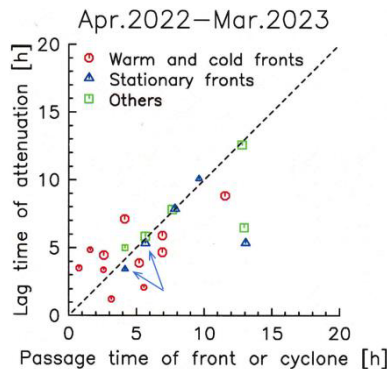


Fig. 4 Scatter plot between the passage time of front or cyclone and the lag time of attenuation obtained between the two stations.

the three stations. Figure 4 shows a scatter plot between the passage time of fronts or cyclones and the lag time of attenuation obtained between Matsuyama and Neyagawa stations (small symbol) and Neyagawa and Yokosuka stations (large symbol). The events are classified into those caused by warm and cold fronts (red circle) and those caused by stationary fronts (blue triangle). The events caused by a cyclone alone without fronts are shown as others (green square). The passage time of warm and cold fronts is measured by the fronts instead of the cyclone center as shown in Fig. 3(b) [10]. In addition, blue arrows indicate the events shown by Figs. 1 and 2. The cross-correlation coefficient between the passage time and the lag time is about 0.67 for all of the rainfall events.

Note that the lag time caused by warm and cold fronts is concentrated around 2–8 h and comparatively shorter than that caused by cyclone passage, because the fronts, in general, pass through the two stations diagonally as shown by Fig. 3(b), and yield shorter distances between the stations perpendicular to the fronts. Thus, the lag time of rain attenuation occurrence between the stations is found to be largely affected by the rain area motions due to rain fronts and cyclones.

4. Duration Time of Rain Attenuation

It should be here noted that the distribution of duration time exceeding specified rain attenuation is important for the practical operation of satellite communications links. The basic statistics of duration time of the rain attenuation were already presented in the reference [2]. In this chapter, however, the characteristics of duration time are further investigated in light of rainfall rate and rain area motion obtained in the previous chapter. Then, the horizontal scale of rain cells expected from the duration time is discussed using the rain area motion.

Figure 5 shows examples of the duration time exceeding 5 dB attenuation observed on August 13, 2022 at Yokosuka station. The duration time (red) and the speed of rain area motion (blue) are averaged over each rainfall event. If the rain attenuation exceeding 5 dB is not observed during more

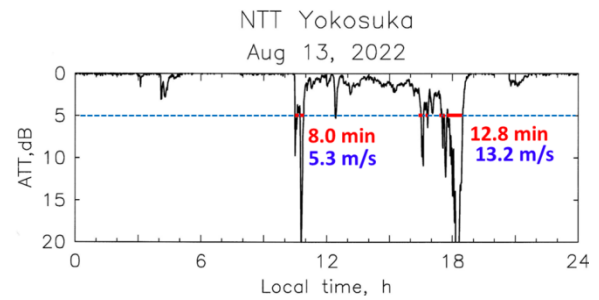


Fig. 5 Examples of the duration time (red) exceeding 5 dB attenuation and the speed of rain area motion (blue) observed on August 13, 2022 at Yokosuka station.

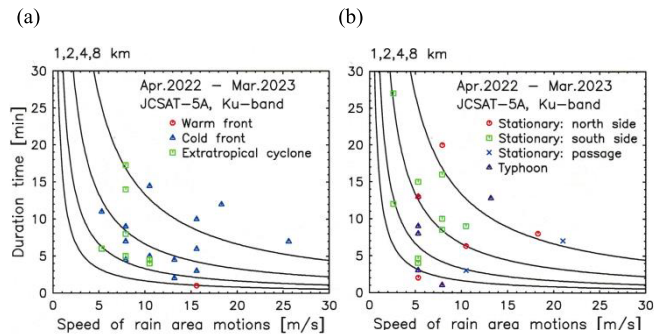


Fig. 6 Duration time of Ku-band JCSAT-5A beacon signal attenuation, and its relationship with the rain area motion for (a) warm or cold fronts, and extratropical cyclone, and (b) stationary fronts and typhoon.

than 3 h, the next rainfall event is regarded as a new event.

Figure 6 shows the duration time of Ku-band JCSAT-5A beacon signal attenuation exceeding 5 dB and its relationship with the speed of rain area motion obtained at the three stations for each rainfall event in the case of the rainfall rate of more than 10 mm/h from Apr. 2022 to Mar. 2023. Figure 6(a) indicates the case of warm or cold fronts and extratropical cyclones, while Fig. 6(b) indicates the case where the attenuation occurs on the north side, the south side, and in the passage of stationary fronts, together with typhoons. The speed of rain area motion is inferred from the weather charts in the same way as shown by the red arrows in Fig. 3.

The thin curves in Fig. 6 show the horizontal scale of rain areas estimated from the product of duration time and rain area passage speed, and this corresponds to the scale of rain cells that produce rain attenuation of 5 dB or more. It is found in Fig. 6 that the duration time is, in general, decreased as the speed of rain area motion is increased, according to the thin curves with each horizontal scale.

The duration time of rain attenuation and the horizontal scale of rain cells associated with 5 dB attenuation are also considered to be largely affected by the average rainfall rate of each rainfall event. Figure 7 shows the scatter plot between the rainfall rate and the horizontal scale inferred from the thin curves in Fig. 6, for (a) warm or cold fronts and cyclones and (b) on the north side, the south side, and in the passage of stationary fronts, together with typhoons.

It is found in Fig. 7 that, as the average rainfall rate is

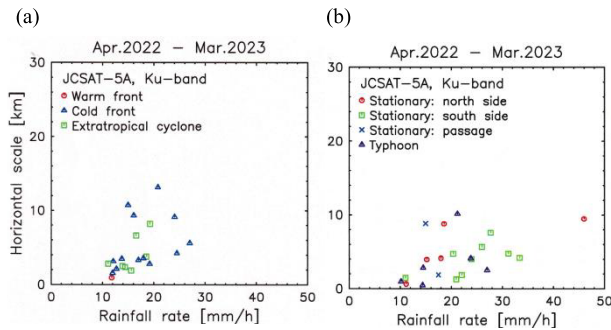


Fig. 7 Scatter plot between the horizontal scale of rain cell and the rainfall rate for each rain type including typhoon.

increased up to about 30–50 mm/h, the horizontal scale of the rain cells is increased for each rainfall type. The maximum horizontal scale of the rain cells is, however, shown to be limited to about 10 km, on an average, in these 43 rainfall events obtained from Apr. 2022 to Mar. 2023 at the three stations. Also, these features are similarly found in each rain type at all of the three stations. Compared to Fig. 4, the number of rainfall events is increased about twice, because the rainfall events are counted even though the rainfall rate of more than 10 mm/h is not measured for longer than 30 min simultaneously at the three stations.

5. Time Delayed Diversity Effects

In this chapter, we will discuss how to deal with the rain attenuation fluctuation in the order of minutes, which is important in actual operation of satellite links. As an example, we introduce time delayed diversity effects that use information signals of two channels in one or two satellites transmitted with a certain time difference. This diversity method is considered to play an important role in the digital satellite links with information storage type communications. Also, we investigate the effects of rain area motion above the observation point on the time delayed diversity.

Figure 8 shows the cumulative time percentages of rain attenuation (thick line) obtained at (a) Neyagawa, (b) Yokosuka, and (c) Matsuyama stations from April 2022 to March 2023. Moreover, the time delayed diversity effects are calculated by these signals measured at 1-min interval, and presented for the lag time of 5 to 40 min in the lower part of the diagrams. The results of numerical calculations are indicated by thin dotted lines with different symbols, respectively.

It can be seen from Fig. 8 that the diversity effects due to the reduction of time percentages by an order or more are available when the lag time of 40 min is provided for the attenuation of 5 dB or more, which may be a considerable problem in the satellite propagation paths. In addition, even with the lag time of only 10 to 20 min, the reduction of time percentages by nearly an order of magnitude is shown to be possible for significant rain attenuation of 10 dB or more with the time percentages of 0.02% or less.

Next, Fig. 9 shows the time delayed diversity effects on

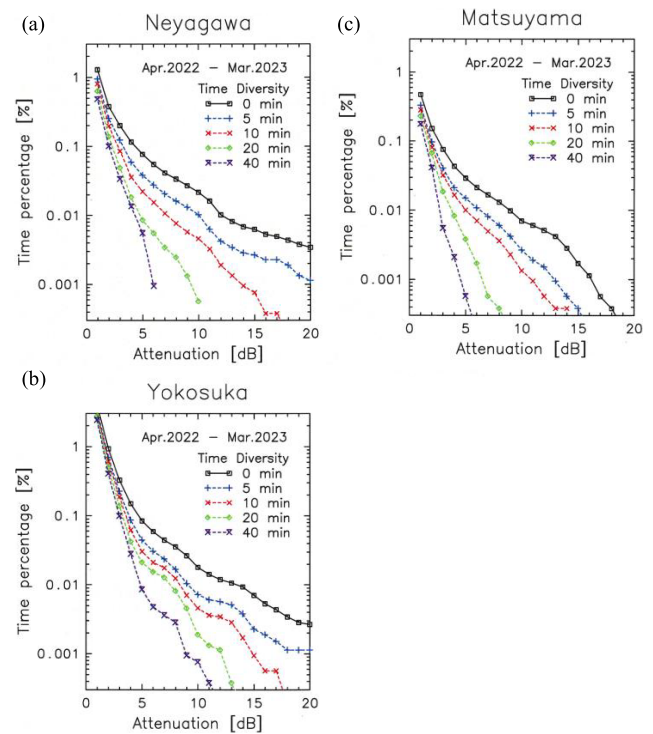


Fig. 8 Cumulative time percentages of rain attenuation obtained (a) Neyagawa, (b) Yokosuka, and (c) Matsuyama stations with the time delayed diversity effects calculated for the lag time from 5 to 40 min.

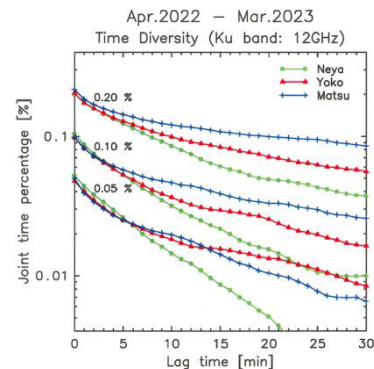
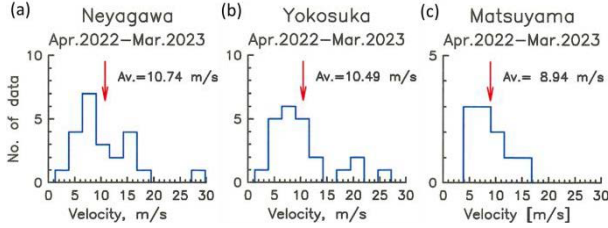


Fig. 9 Time delayed diversity effects on the joint time percentages at the three stations, depending on the lag time from 0 to 30 min.

the joint time percentages at the three stations, depending on the lag time from 0 to 30 min in the horizontal axis. The results are indicated for the time percentages of 0.2, 0.1, and 0.05% in the original cumulative attenuation, respectively, during the one year. The rain attenuation values corresponding to these time percentages at each station are summarized in Table 2. Thus, the joint time percentages are similarly decreased, as the lag time is increased up to 30 min for all stations. However, the time delayed diversity effects are found to be considerably different according to the stations: larger effects by faster decrease in time percentages with lag time are, as a whole, shown at Neyagawa station (green), and smaller effects by slower decrease are shown at Matsuyama

Table 2 Rain attenuation for time percentages at each station.

Time percentage	Neyagawa	Yokosuka	Matsuyama
0.2%	3.0 dB	3.7 dB	1.8 dB
0.1%	4.3 dB	4.8 dB	2.7 dB
0.05%	6.2 dB	6.8 dB	3.8 dB

**Fig. 10** Distribution of the speeds of rain area motion at the three stations.

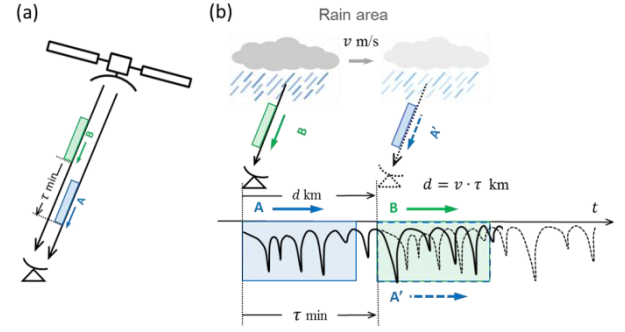
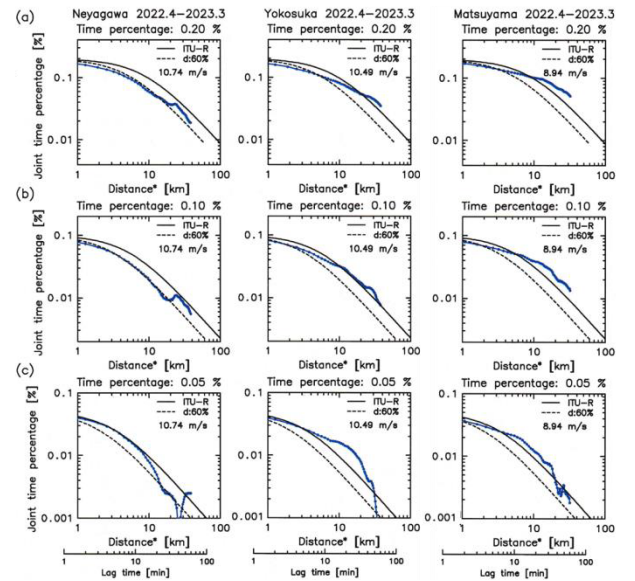
station (blue).

Figure 10 shows the distribution of the speeds of rain area motion for the rainfall events in which the rainfall rate of more than 10 mm/h is recorded for more than 30 min during one year. The speed of rain area motion is similarly inferred from the weather charts as was shown in Fig. 3. The number of events for Neyagawa, Yokosuka and Matsuyama stations is 23, 23, and 10, respectively. Note that the total number of events is 56, and increased more than that of 43 for the duration time in Figs. 6 and 7, because the events with rain attenuation not exceeding 5 dB are included. On the other hand, when the rainfall rate is less than 10 mm/h, however, the attenuation of more than 2–3 dB rarely appears. So, this case does not affect the attenuation statistics with time percentages of less than 0.2% as shown in Table 2.

It should be noted in Fig. 10 that the speed of rain area motion is comparatively larger at Neyagawa station which indicates the faster decrease in time percentages in Fig. 9, while it is rather smaller at Matsuyama station which indicates the slower decrease in time percentages in Fig. 9. Thus, the speed of rain area motion is considered to affect the time delayed diversity effects at each station.

6. Conversion to Site Diversity Effects

Figure 11(a) illustrates the concept of time delayed diversity and (b) the relationship between the time delayed diversity and the site diversity, respectively. The time delayed diversity is conducted by comparing the information of the channel A with that of the channel B which is sent and received later by the lag time of τ min. In Fig. 11(b), the information received earlier and stored during the lag time of τ min is denoted by A' . When the rain area moves the distance d km with the speed of rain area motion v m/s, keeping its overall configuration, the time delayed diversity is shown to be equivalent to the site diversity with the distance d km. That is, the diversity is considered to be conducted by comparing the channel B with the channel A' received at a virtual earth station with distance d km. In this way, we can propose a method to convert the time delayed diversity effect measured at a single station to the site diversity effect for the distance

**Fig. 11** (a) Concept of time delayed diversity and (b) relationship between the time delayed diversity and the site diversity.**Fig. 12** Joint time percentages obtained from the time delayed diversity with the lag time of 1 to 60 min, using the rain attenuation of Ku-band signals measured at the three stations.

d km around the station using the speed of rain area motion v m/s.

Next, Fig. 12 indicates the joint time percentages of the time delayed diversity at the three stations shown in Fig. 9 again, with the lag time up to 60 min. The results are indicated for the original time percentages of (a) 0.2, (b) 0.1, and (c) 0.05%, respectively, corresponding to the attenuation listed in Table 2 at each station. The horizontal axis is here converted from the lag time to the distance of rain area motion at each station, by multiplying the average speed of the rain area motion shown in Fig. 10. Also, the distance is indicated by a logarithmic scale.

Then, the joint time percentages are compared with the predicted values of the site diversity effect by the ITU-R recommendations [11]. The solid line is the conventional ITU-R recommendations when the moving direction of the rain area is random with respect to the direction between the two stations performing the site diversity. On the other hand, the dotted line is the case that the improvement rate is the

maximum when the direction of rain area motion coincides with the direction between the two stations performing the site diversity. It is known that in this case, the distance between the two stations becomes about 60% of the ITU-R recommendations [10].

In Fig. 12, the joint time percentages tend to approach the dotted line, when the distance of rain area motion is shorter than 10 km. Here, the dotted line means that the direction of rain area motion matches the direction between the two stations as shown in Fig. 11. As the distance becomes around 10 km or more, however, each joint time percentage approaches the solid line and becomes slightly larger. Consequently, the equivalent site diversity effect associated with the rain area motion, as a whole, becomes described by the ITU-R recommendations fairly well. As a result, It can be seen that the time delay diversity with a delay time of about 15 min produces the site diversity effect equivalent to a distance of about 10 km.

Note that the distance d km is here along the direction of rain area motion. So this substantial distance, so to speak, becomes at first about 60% of the ITU-R recommendations in the short distance as shown by the dotted line in Fig. 12, due to the almost same rain area configuration. As the distance is increased to more than 10 km with the increase of the lag time, however, the actual rain area configuration may begin to change slightly. Thus, the correlation of rain attenuation is rather reduced in more than 10 km. This may cause the increase in actual site diversity effects, resulting in lower time percentages of the ITU-R predictions shown by the thin lines. Eventually, the joint time percentages of the virtual site diversity that just assumes the same rain area configuration as in the time delayed diversity seem to indicate slightly larger time percentages. Thus, the joint time percentages rather agree well with the ITU-R predictions. Also, this feature seems to be concerned with the fact that the maximum horizontal scale of rain cells is about 10 km in Fig. 7.

Next, Fig. 13 depicts the time delayed diversity effects on the joint time percentages simultaneously at Neyagawa, Yokosuka, and Matsuyama stations, for the lag time up to 30 min and the time percentages of 0.2, 0.1, and 0.05%, respectively. The horizontal axis is here converted to the virtual distance corresponding to the product of the lag time and the rain area motion, as illustrated in Fig. 11. Also, the thin lines are the ITU-R recommendations of the site diversity effect in this distance.

After the conversion, the joint time percentages of the three stations are found to agree well with each other, up to the virtual distance of nearly 10 km in spite of the difference in the speeds of rain area motion. This is also consistent with the maximum horizontal scale of about 10 km in Fig. 7. Thus, these results indicate the effects of the rain area motion to the time delayed diversity technique, and strongly support that the time delayed diversity effects should be equivalently treated as the site diversity effects using the rain area motion.

It should be here noted that the rain attenuation statistics may be largely affected by the local climate and the seasonal variation or type of rainfall, such as typhoon, sum-

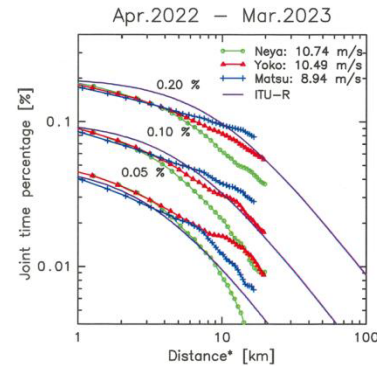


Fig. 13 Time delayed diversity effects on the joint time percentages at Neyagawa, Yokosuka, and Matsuyama stations for the time percentages of 0.2, 0.1, and 0.05%. The horizontal axis is converted to the virtual distance corresponding to the rain area motion.

mer time shower, and rainy seasons. Also, the features of joint time percentages are quite different depending on the type of each rainfall event, as well as the speed of rain area motion. Therefore, the rain attenuation data measured at 1-min interval should be accumulated during at least one year, and averaged over the number of rainfall events presented in Fig. 10, such as 23, 23, and 10 times for Neyagawa, Yokosuka and Matsuyama stations, respectively. As a result, the joint time percentages can be successfully converted from the time delayed diversity to the site diversity up to nearly 10 km.

7. Comparison between Time Delayed Diversity and Site Diversity Effects

This phenomenon is also confirmed by the previous study on the site diversity effects around OECU [10]. That is, the rain area motions causing rain attenuation were estimated by two kinds of BS signal three-point observations from 2002 to 2012. One of the three points in the wide area of 20–50 km consists of Osaka Electro-Communication University (OECU, Neyagawa, Osaka), Kyoto University (RISH, Uji, Kyoto), and Shigaraki MU radar (MU, Koka, Shiga). The other of the three points in the narrow area of 4–8 km consists of Osaka Electro-Communication University (OECU, Neya) and two other premises of our university at Moriguchi (Mori) and Shijonawate (Shijo) cities. The locations of these three-point BS signal observations in the wide and narrow areas are illustrated in Fig. 14 [12].

Figure 15 shows the cross-correlation coefficients of the attenuation obtained from six pairs of the BS signal three-point observations with the distances from about 4 to 46 km. The three of the six pairs in the narrow area are composed of Neya-Shijo (3.87 km), Neya-Mori (5.57 km), and Mori-Shijo (8.19 km), while others in the wide area are OECU-RISH (23.3 km), RISH-MU (28.5 km), and OECU-MU (45.9) as shown in Fig. 14. The average (circle) and standard deviation (vertical bar) of the cross-correlation coefficients are calculated for more than 80 rainfall events when the rain attenuation exceeding 5 dB occurred in each pair of two sta-

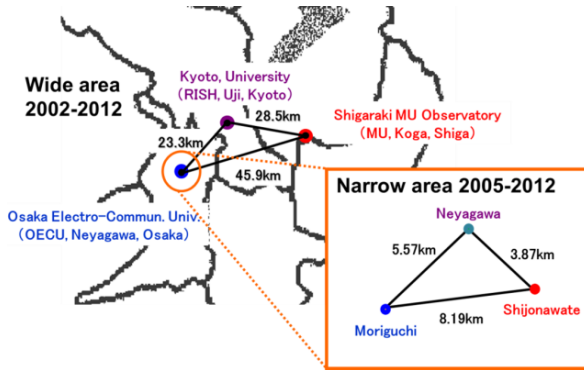


Fig. 14 Locations of wide and narrow area three-point observations of the BS signals around OECU [12].

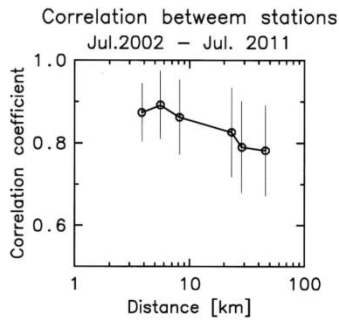


Fig. 15 Cross-correlation coefficients obtained from six pairs of the BS signal three-point observations with the distances from 4 to 46 km.

tions for both narrow and wide areas. It should be noted that the cross-correlation coefficients of nearly 0.9 are maintained in the three pairs of the narrow area with the distances of less than 10 km, while the coefficients are reduced down to around 0.8 in those of the wide area with the distance of more than 10 km.

Next, the time delayed diversity effects are investigated at the six stations in these narrow and wide areas. The joint time percentages averaged for the three pairs in the narrow area are shown in Fig. 16 for the original time percentages of (a) 0.2, (b) 0.1, and (c) 0.05% with the lag time up to 60 min. The horizontal axis is also converted from the lag time to the distance of rain area motion using the average speed of 10.5 m/s in this area. The vertical dotted lines indicate the distance between each pair of the stations. Similarly, the joint time percentages approach the dotted line in the distance of shorter than 10 km, while they approach the solid line and become slightly larger. Thus, it is found that the equivalent site diversity effects in the narrow area can be described by the ITU-R recommendations with the distance up to about 30 km.

Furthermore, the joint time percentages averaged for the three pairs in the wide area are similarly shown in Fig. 17 for the original time percentages of (a) 0.2, (b) 0.1, and (c) 0.05% with the lag time up to 90 min. The distance of rain area motion is similarly estimated using the average speed of 10.5 m/s. Thus, the distance dependency of the joint time

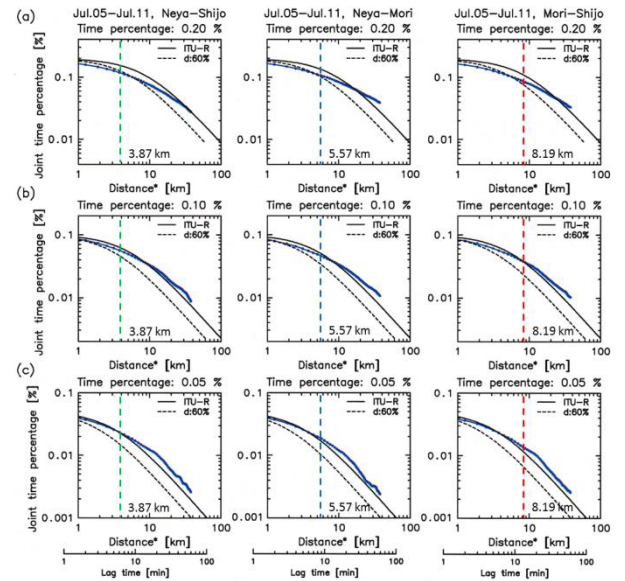


Fig. 16 Joint time percentages obtained from the time delayed diversity with the lag time of 1 to 60 min, using the rain attenuation of Ku-band BS signals averaged for three pairs in the narrow area. The vertical dotted lines indicate the distance between each pair.

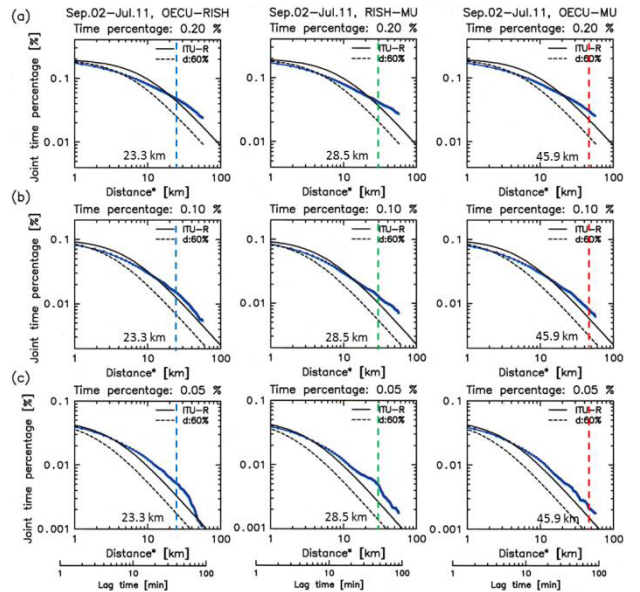


Fig. 17 Joint time percentages obtained from the time delayed diversity with the lag time of 1 to 90 min, using the rain attenuation of Ku-band BS signals averaged for three pairs in the wide area. The vertical dotted lines indicate the distance between each pair.

percentages in the wide area shows much the same features as in the narrow area, and the equivalent site diversity effects described by the ITU-R recommendations with the distance up to about 50 km. Thus, it is found that the site diversity effects are successfully estimated for the virtual distances of about 30 km and 50 km in the narrow and wide areas, respectively, compared with about 10 km in the present three stations in Fig. 12. The extension of the distance seems to be

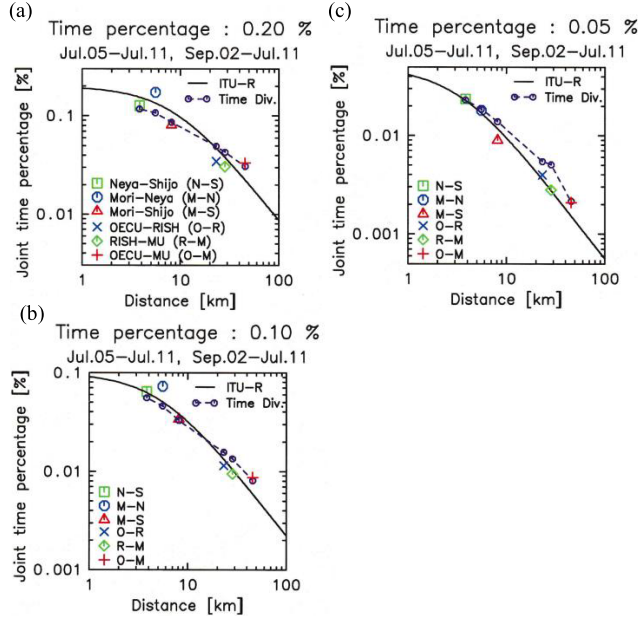


Fig. 18 Site diversity effects calculated at the virtual distance from the time delayed diversity (dotted line) and those obtained from the direct measurements at the six pairs of the BS signal three-point observations.

caused by the longer observation periods of about 5 and 10 years in these areas, while the period is only one year in the present observation. Note that the longer periods increase the statistical reliability of the joint time percentages used in the conversion from time delayed diversity to site diversity.

Finally, the site diversity effects calculated at the virtual distance from the time delayed diversity are compared with those actually obtained from the direct measurements using the six pairs of the BS signal three-point observations in the narrow and wide areas, respectively [12]. The time percentages of the time delayed diversity are obtained from the average of each pair of the BS signal observations in the corresponding distances as shown by the vertical dotted lines in Figs. 16 and 17. The results are shown in Fig. 18 for both calculations (dotted line) and measurements (each symbol) of the site diversity effects in the time percentages of (a) 0.2, (b) 0.1, and (c) 0.05%, as well as the ITU-R recommendations (solid line).

Thus, the site diversity effects inferred from the time delayed diversity are found to agree well with the actual observations in each distance of the BS signal three-point observations of the narrow area up to 10 km. In the wide area of more than 10 km, however, the time percentages are slightly larger than those of the actual observations. These discrepancies seem to be caused by the decrease in the cross-correlation coefficients of the actually observed rain attenuation between the two stations, as shown in Fig. 15 in the distance of longer than 10 km. However, the discrepancies are not so significant, but the ITU-R predictions are still considered to be effective. These results strongly support the validity of equivalence of the time delayed diversity to the site diversity using the rain area motion as illustrated in

Fig. 11.

As a result, the site diversity effects inferred from the time delayed diversity at Neyagawa, Yokosuka, and Matsuyama stations also seem to be very reliable at least up to the distance of about 10 km and for the original time percentage of more than 0.05% as shown in Fig. 13.

This novel method to convert the time delayed diversity to the site diversity at only one station is considered to be very useful. In this method, the rain attenuation measurements at 1 min interval during one year can even predict the site diversity effects for the nearby stations up to at least 10 km distance, using the rain area motion obtained from weather charts. Moreover, this information will be important for the future NTN, because a number of mobile stations located at nearby distances may simultaneously access LEO or HAPS. Note that fixed satellite services based on GEO usually conduct the site diversity at the distance of about 20–50 km [12], while the mobile stations in the NTN may exist within the distance of 10 km each other. Therefore, the diversity effects of about 10 km distance obtained in this study seem to be useful to estimate the efficiency of the network operation among these mobile stations.

In the distance of longer than 10 km, however, the diversity effects show considerable discrepancies among the three stations in Fig. 13, compared to the previous BS signal observations in Fig. 18. To confirm the diversity effects in the longer distance, further observation is required in more than one year, as shown by the past three-point BS signal observations.

8. Conclusions

The characteristics of rain attenuation statistics for the Ku-band satellite signals in relation to the rain area motion are investigated among the three earth stations at Osaka Electro-Communication University (OECU) Neyagawa, NTT Yokosuka R&D Center (Yokosuka, Kanagawa), and satellite base station (Matsuyama, Ehime), respectively, from April 2022 to March 2023. The time difference of the attenuation occurrence among these stations is well explained by the rain area motion associated with rain fronts and extratropical cyclones obtained from the weather charts. Also, the duration time of the attenuation are shown to be largely affected by the speed of the rain area motion around each station. The scale of rain cells inferred from the product of duration time and rain area speed is found to be increased up to around 10 km at most for the attenuation exceeding 5 dB, as the average rainfall rate is increased at each rainfall event.

We investigate the time delayed diversity effects as a method to deal with the rain attenuation fluctuation in the order of min, based on the 1-min measurements at each station. Then, we proposed a new conversion method to estimate the site diversity effects from attenuation data observed at only one station during one year, using the speed of rain area motion obtained from the weather charts near each observation point. As a result, the site diversity effects can be similarly estimated at each station up to about 10 km

from time delayed diversity of about 15 min. Furthermore, these joint time percentages agree fairly well with the ITU-R recommendations, demonstrating the effectiveness of this conversion method for distances up to 10 km. This result is also considered to be very useful for future NTN operations, in which a number of mobile stations located at nearby distances may simultaneously access LEO or HAPS instead of GEO.

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